Mohimenul Kabir (Mahi)

PhD Candidate, National University of Singapore
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Research interests

Automated Reasoning, Formal Methods, Logic Programming, Answer Set Programming, Artificial Intelligence, Metaheuristic

Education

2020 – **Present** **National University of Singapore** (NUS) – Singapore
PhD in Computer Science
Advisor: Kuldeep S. Meel.

2013 – 2017 **Bangladesh University of Engineering and Technology** (BUET) – Bangladesh
BSc in Computer Science
Mentors: M Sohel Rahman.

Honors and scholarships

2023 NUS SoC Research Incentive Award

2020 NUS Research Scholarship

2014 & 2015 University Merit Scholarship (Bangladesh University of Engineering and Technology)

Publications

2024 **On Lower Bounding Minimal Model Count**
Best Paper award of ICLP 2024
Mohimenul Kabir and Kuldeep S Meel
Theory and Practice of Logic Programming (Special issue of ICLP 2024).

2024 **Exact Answer Set Counting with Compact Encoding**
Mohimenul Kabir, Supratik Chakraborty and Kuldeep S Meel
Proceedings of the AAAI Conference on Artificial Intelligence 2024 (**Invited in Model Counting and Sampling Workshop 2024**).

2023 **A Fast and Accurate ASP Counting Based Network Reliability Estimator**
Mohimenul Kabir and Kuldeep S Meel
International Conference on Logic for Programming, Artificial Intelligence and Reasoning (LPAR) 2023. (**Invited in Model Counting Competition Workshop 2023**)

2023 **Multi-objective optimization and heuristic based solutions for evacuation modeling**
Mohimenul Kabir, Jaiaid Mobin, Muhammad Ali Nayeem, Muhammad Ahsanul Habib, and M. Sohel Rahman
Transportation Research Interdisciplinary Perspectives.

2022 **ApproxASP – A Scalable Approximate Answer Set Counter**
Mohimenul Kabir, Flavio O Everardo, Ankit K Shukla, Markus Hecher, Johannes Klaus Fichte, and Kuldeep S Meel
Proceedings of the AAAI Conference on Artificial Intelligence 2022. (**Invited in ICLP 2022 and Indian SAT-SMT school 2022**)

2020 **Evaluation of classification and forecasting methods on time series gene expression data**
Nafis Irtiza Tripto, **Mohimenul Kabir**, Md Shamsuzzoha Bayzid, Atif Rahman
PLOS ONE.

2020 **ADACT: a tool for analysing (dis) similarity among nucleotide and protein sequences using minimal and relative absent words**
Mujtahid Akon, Muntashir Akon, **Mohimenul Kabir**, M Saifur Rahman, M Sohel Rahman
Bioinformatics.

2018 **A simple, fast, filter-based algorithm for circular sequence comparison**
Md Azim, Aashikur Rahman, **Mohimenul Kabir**, M Sohel Rahman
In International Workshop on Algorithms and Computation (WALCOM).

Research Visits

Oct 2024 **Texas Tech University**

Jan – June 2024 **University of Toronto**

Teaching experience

Fall 2023 **Teaching assistant, IT5005: Introduction to Artificial Intelligence (NUS)**

Spring 2021 & 2023 **Teaching assistant, CS4244: Knowledge Representation and Reasoning (NUS)**

Fall 2021 **Teaching assistant, CS3243: Introduction to Artificial Intelligence (NUS)**

Spring 2022 **Teaching assistant, CS4246/CS5446: AI Planning and Decision Making (NUS)**

Fall 2022 **Teaching assistant, CS2040: Data Structures and Algorithms (NUS)**

Talks

October 2024 On Lower Bounding Minimal Model Count
ICLP 2024

October 2024 Answer Set Counting and its Applications
ICLP LPNMR Doctoral Consortium 2024

August 2024 Exact ASP Counting with Compact Encodings
Model Counting and Sampling workshop 2024

February 2024 Exact ASP Counting with Compact Encodings
AAAI 2024

June 2023 — A Fast and Accurate ASP Counting Based Network Reliability Estimator
LPAR 2023

December 2022 — ApproxASP – a scalable approximate answer set counter
SAT SMT Winter School 2022

February 2022 — ApproxASP – a scalable approximate answer set counter
AAAI 2022

Mentorship and service

May – July 2022 — **National University of Singapore (Academic Advisor)**
My responsibilities: I mentored two undergraduate students on a research project. The title of the project: "Approximate Answer Set Sampling"

Technical skills

Programming languages: C, C++, Java, Python, R

References

Kuldeep S Meel (meel@cs.toronto.edu, University of Toronto)

Supratik Chakraborty (supratik@cse.iitb.ac.in, IIT Bombay)

Johannes K. Fichte (johannes.fichte@gmail.com, Linkoping University)